Project: Worked examples
Title: Thick-walled cylinder stresses (Lamé)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Radial, hoop and axial stresses at the bore and the outer surface of a thick-walled cylinder under internal and external pressure, the von Mises and Tresca equivalents, utilisation against an allowable, and the thin-wall estimate for comparison.

WHAT IT CALCULATES	RELATION
Lamé constants from the bore and outside pressures	$A = (a^2 p_i - b^2 p_o)/(b^2 - a^2)$; $B = a^2 b^2 (p_i - p_o)/(b^2 - a^2)$
Radial and hoop stress at radius r	$\sigma_r = A - B/r^2$; $\sigma_\theta = A + B/r^2$
Axial stress, closed or open ends	$\sigma_z = A$ (closed); $\sigma_z = 0$ (open)
Equivalent stresses from the three principal stresses	von Mises; Tresca = the largest principal difference

Basis: Thick-walled cylinders[1](Ch. 13, Table 13.5); Stresses in pressurised cylinders[2](§3-14).

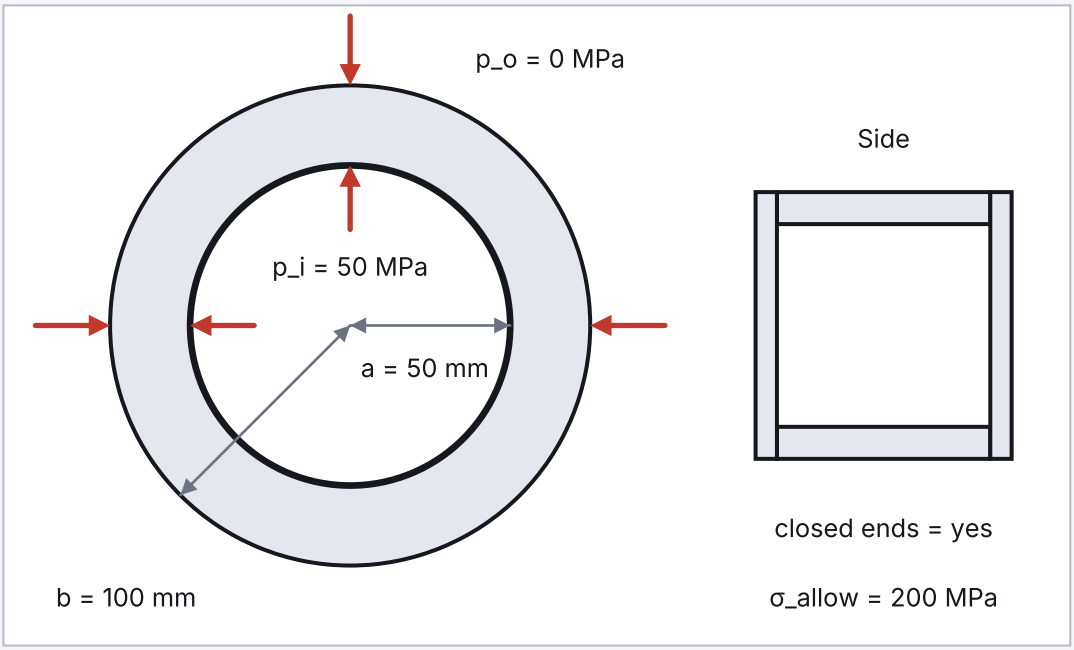


Figure 1 — A thick-walled cylinder in section, bore radius a, outer radius b, pressure inside and out; closed ends carry the pressures axially. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
a	Inner radius	50	mm	drawing: DRG-CYL-01 rev A	0.1 … 5000 mm
b	Outer radius	100	mm	drawing: DRG-CYL-01 rev A	0.1 … 5000 mm
p_i	Internal pressure	50	MPa	spec: SPEC-PV-2 §3 design pressure	0 … 1000 MPa

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
p_o	External pressure	0	MPa	assumption: Atmospheric outside.	0 … 1000 MPa
closed	Closed ends (axial stress carried by the wall)	yes		drawing: DRG-CYL-01 end caps rev A	any
sigma_allow	Allowable stress	200	MPa	spec: SPEC-PV-2 §5	1 … 3000 MPa

3. Assumptions

A1. Long cylinder, elastic, isotropic; stresses vary with radius only. (Lamé solution; end effects neglected.)
A2. Closed ends carry the pressure end load as a uniform axial stress. (σ_z = A.)
A3. p_o (External pressure) is taken as 0 MPa. (Atmospheric outside.)

4. Working

[7] A = (a^2 * p_i - b^2 * p_o) / (b^2 - a^2) [1](Table 13.5)
= ((50 mm)^2 × 50 MPa − (100 mm)^2 × 0 MPa) / ((100 mm)^2 − (50 mm)^2)
= 16.667 MPa
Lamé constant A

[8] B = a^2 * b^2 * (p_i - p_o) / (b^2 - a^2) [1](Table 13.5)
= (50 mm)^2 × (100 mm)^2 × (50 MPa − 0 MPa) / ((100 mm)^2 − (50 mm)^2)
= 166670 MPa·mm²
Lamé constant B

[9] sigma_r_a = A - B / a^2
= 16.667 MPa − 166670 MPa·mm² / (50 mm)^2
= -50 MPa
Radial stress at the bore

[10] sigma_t_a = A + B / a^2
= 16.667 MPa + 166670 MPa·mm² / (50 mm)^2
= 83.333 MPa
Hoop stress at the bore

[11] sigma_r_b = A - B / b^2
= 16.667 MPa − 166670 MPa·mm² / (100 mm)^2
= 1.8626e-15 MPa
Radial stress at the outer surface

[12] sigma_t_b = A + B / b^2
= 16.667 MPa + 166670 MPa·mm² / (100 mm)^2
= 33.333 MPa
Hoop stress at the outer surface

[13] sigma_z = A
= 16.667 MPa
= 16.667 MPa
Axial stress (closed ends)

[14] sigma_vm_a = sqrt(((sigma_t_a - sigma_r_a)^2 + (sigma_r_a - sigma_z)^2 + (sigma_z - sigm
2)
= sqrt(((83.333 MPa − (-50 MPa))^2 + ((-50 MPa) − 16.667 MPa)^2 + (16.667 MPa − 83.333 MPa)^2
= 115.47 MPa
von Mises stress at the bore

[15] sigma_vm_b = sqrt(((sigma_t_b - sigma_r_b)^2 + (sigma_r_b - sigma_z)^2 + (sigma_z - sigm
2)
= sqrt(((33.333 MPa − 1.8626e-15 MPa)^2 + (1.8626e-15 MPa − 16.667 MPa)^2 + (16.667 MPa − 33.
MPa)^2) / 2)
= 28.868 MPa

von Mises stress at the outer surface

$$[16]\ \sigma_{tr,a} = \max(\sigma_{t,a}, \sigma_{r,a}, \sigma_z) - \min(\sigma_{t,a}, \sigma_{r,a}, \sigma_z)$$
$$= \max(83.333\ \text{MPa}, -50\ \text{MPa}, 16.667\ \text{MPa}) - \min(83.333\ \text{MPa}, -50\ \text{MPa}, 16.667\ \text{MPa})$$
$$= 133.33\ \text{MPa}$$

Tresca stress at the bore

$$[17]\ \sigma_{tr,b} = \max(\sigma_{t,b}, \sigma_{r,b}, \sigma_z) - \min(\sigma_{t,b}, \sigma_{r,b}, \sigma_z)$$
$$= \max(33.333\ \text{MPa}, 1.8626\text{e-}15\ \text{MPa}, 16.667\ \text{MPa}) - \min(33.333\ \text{MPa}, 1.8626\text{e-}15\ \text{MPa}, 16.667\ \text{MPa})$$
$$= 33.333\ \text{MPa}$$

Tresca stress at the outer surface

$$[18]\ \sigma_{vm,max} = \max(\sigma_{vm,a}, \sigma_{vm,b})$$
$$= \max(115.47\ \text{MPa}, 28.868\ \text{MPa})$$
$$= 115.47\ \text{MPa}$$

Maximum von Mises stress

$$[19]\ u = \sigma_{vm,max} / \sigma_{allow}$$
$$= 115.47\ \text{MPa} / 200\ \text{MPa}$$
$$= 0.57735$$

Utilisation

$$[20]\ \sigma_{thin} = (p_i - p_o) * (a + b) / 2 / (b - a)$$
$$= (50\ \text{MPa} - 0\ \text{MPa}) \times (50\ \text{mm} + 100\ \text{mm}) / 2 / (100\ \text{mm} - 50\ \text{mm})$$
$$= 75\ \text{MPa}$$

Thin-wall hoop estimate at the mean radius

$$[21]\ k_{wall} = b / a$$
$$= 100\ \text{mm} / 50\ \text{mm}$$
$$= 2$$

Wall ratio b/a

5. Results

SYMBOL	VALUE	DESCRIPTION
sigma_r_a	-50 MPa	Radial stress, bore
sigma_t_a	83.333 MPa	Hoop stress, bore
sigma_r_b	1.8626e-15 MPa	Radial stress, outer
sigma_t_b	33.333 MPa	Hoop stress, outer
sigma_z	16.667 MPa	Axial stress
sigma_vm_a	115.47 MPa	von Mises, bore
sigma_vm_b	28.868 MPa	von Mises, outer
sigma_tr_a	133.33 MPa	Tresca, bore
sigma_tr_b	33.333 MPa	Tresca, outer
sigma_vm_max	115.47 MPa	Maximum von Mises
u	0.57735	Utilisation
sigma_thin	75 MPa	Thin-wall hoop estimate
k_wall	2	Wall ratio

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	von Mises against allowable	u = 0.57735	u ≤ 1	0.577	

7. Applicability

No warnings.

8. Verification

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Template mech.thick-cylinder@2.0.1 Content hash faff1841ff151c3a Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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9. References

1. Young WC, Budynas RG, Sadegh AM. Roark's Formulas for Stress and Strain. 8th ed. McGraw-Hill; 2012.
2. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.